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Solar pyrolysis of waste plastics with photothermal catalysts for high-value products

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ABSTRACT

The photothermal catalytic pyrolysis of low-density polyethylene (LDPE) plastics has been investigated with bifunctional photo-thermo catalysts in a novel photothermal reaction system. Six kinds of catalysts were introduced and explored in an in-situ fixed-bed reactor under 500 °C heated by solar simulator. The results indicated that the hydrogen production and jet fuel selectivity of Ni-Ti-Al catalyst were 34 mol/kg and 80% respectively, which were significantly higher than 7 mol/kg and 38% of Ti-Al. Catalysts loaded with Ni exhibited strong selectivity for aromatic hydrocarbon, achieving 49.51% for Ni-Al and 53.46% for Ni-Ti-Al especially. In addition, Ni-Ti-Al led to high-quality filamentous carbon tubes in smaller diameter, which presented high thermal stability and graphitization. Bifunctional catalyst, especially Ni-Ti-Al possesses obvious photothermal catalytic performance, which is ascribed to relaxations of photogenerated inter- and intra-band electrons of plasmonic metal Ni and excellent carriers transfer capability of non-plasmonic semiconductor TiO_2. The production of hydrogen and jet fuel from plastic by photothermal catalytic pyrolysis provides an environmentally friendly way for high-value-recycling of large amount of the waste plastics.

1. Introduction

Plastic materials have been widely used with a huge consumption on a daily basis due to their excellent properties such as light weight, low cost, versatility or good insulation [1]. The world plastic production reached 368 million tonnes in 2019, whereas there is nearly 25% of post-consumer waste sent into landfill [2,3]. The circular economy of plastic is calling for zero landfilling, while recycling is the first option to close the loop for waste disposal. Recently, the technology of "turning waste into treasure", such as converting waste plastics into chemicals or high value-added products, prevails in the research of waste treatment, among which pyrolysis is one of the most promising technologies [4–6]. Valuable gas, solid and liquid products can be obtained from plastic chemical conversion. The obtained liquid oil from plastics is expected to be an alternative energy source for automobile fuel due to its similar combustion properties with diesel and gasoline, and the solid carbon can be disposed to obtain high-quality carbon nanotubes [5].

The introduction of catalyst into plastic pyrolysis can significantly shorten the reaction time, reduce the reaction temperature and increase the conversion rate, while the product selectivity and distribution can be tuned by different catalysts [7,8]. Featured with high mechanical strength, rich Lewis acid sites as well as porous structure, γ-Al_2O_3 is widely used as catalyst or catalyst support material for thermal decomposition [9–11]. Sakashita et al. [12] introduced Al_2O_3 as a substrate into fluid catalytic cracking (FCC) and found that it could not only obtain high activity, but also reduce coking. Marczewski et al. [13] studied the influence of the acid and basic properties of γ-Al_2O_3 on the selectivity of polystyrene transformation. Shen et al. [14] found that the alumina component and its interaction with Ni significantly enhanced the product yield and graphitization of solid product. Celikgogus and Karaduman. [15] investigated the effects of γ-Al_2O_3 loaded with different metals (Ni, Cu, Ce, Co) and reaction temperature on the

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pyrolysis of polystyrene waste foams and obtained the highest styrene yields over Cu/γ-Al_2O_3 at 500 °C.

Plastic pyrolysis is mainly thermal chemical conversion usually accompanied by high electricity or fuel consumption thus making the technology less efficient and environmental-friendly. Photocatalytic degradation of solid waste provides a new option for waste treatment, where the clean, cheap and pollution-free sunlight can be utilized as a driving energy [16]. As one kind of traditional semiconductor, TiO_2 is the most widely applied photocatalyst due to the characteristics such as inexpensiveness, high stability, non-toxicity, high-reactivity and so on [17]. Moreover, it can absorb visible light or even near infrared light (NIR) by element doping and introduction of oxygen vacancies, which effectively enhance carriers transport and shift the Femi level of TiO_2 toward the conduction band to promote the carrier separation [18–20]. Graphitic-carbon nitride (g-C_3N_4), with the band gap of 2.7 eV, is a new photocatalyst material with a two-dimensional network structure. Park et al. [21] investigated synergetic effects of g-C_3N_4 supported on γ-Al_2O_3 (Al-CN) for the removal of naphthenic acids in highly acidic model crude oil. They found that the high dispersion of Al-CN improved catalytic activity by adjusting the acidic and basic sites on the bifunctional Al-CN and the alleviation of coke deposition.

Even so, photocatalysis is confronted with many intractable problems, including low conversion efficiency, low energy density, slow reaction rate and so on. Meantime, the development of thermo-catalytic conversion is hindered by its high energy consumption. Such facts motivate the application of the integrated photothermal catalysis for waste disposal in current work. Recently, researchers has found that the introduction of heat into photocatalytic system can reduce above defects, achieving enhanced photocatalytic performance [22]. Photo-thermo-catalysis fulfill the overall solar spectrum utilization including the UV–vis spectra with higher energy to trigger photocatalysis and the heating effect of NIR spectrum which provides the essential thermal energy for endothermic reaction [17,23,24]. When UV–vis light was absorbed by TiO_2, it exhibited an electron jump between valence band and conduction band, thus leaving a positively charged hole, which initiated direct oxidation of intermediate [25]. However, the carriers recombination acted as retardant to the photocatalytic activity of TiO_2, causing an energy waste step [26]. Transition metal and metal oxide doping can increase the catalytic activity through reducing the recombination rate of photo-generated carriers. Peh et al. [27] designed simple reactor with a parabolic reflector, gathering sunlight onto a borosilicate round-bottomed flask containing CuO/TiO_2 for H_2 generation. Nakano et al. [28] prepared Pt/TiO_2-SiO_2 composite catalyst which showed good catalytic performance in the photo-thermal catalytic degradation of acetaldehyde. Photo-thermo catalytic pyrolysis can only be carried out in direct heating solar reactor. The reactants are heated directly by concentrated solar radiation through transparent window. The selection of solar energy resource-rich region is a prerequisite for the application of photo-thermo catalytic pyrolysis technology.

To our best knowledge, it is the first time that plastic waste was recycled by photothermal catalytic pyrolysis using concentrated solar energy. However, the photo and thermal effect in the catalytic conversion of waste plastic remains unclear, and the potential synergistic mechanism between them need further exploration. In this work, a series of photothermal catalysts, including Ti-Al and CN-Al were prepared for improving the quality of products and exploring the mechanism. In addition, the introduction of Ni in this reaction system was further investigated in relation to product distribution and selectivity. This work would provide a feasible and efficient solution for high-value recycling of waste plastics in photothermal catalysis way.

2. Material and methods

2.1. Plastic feedstock and catalyst preparation

Low density polyethylene (LDPE), a typical plastic component, obtained from Hua Chuang Plastic Co. Ltd. (China) was used as feedstock, and it was in the form of powder with particle size about 50 mesh. The ultimate and proximate analyses of LDPE were shown in Table 1 and it can be seen that LDPE had a high content of volatiles and hydrogen. The volatiles were beneficial for the production of pyrolysis oil and the hydrogen was able to provide large amount of hydrogen atoms for the production of H_2. All the chemical precursors for catalyst synthesis were supplied by Aladdin Biochemical Technology Co. Ltd. unless mentioned particularly.

The preparation processes of TiO_2-γ-Al_2O_3 (Ti-Al) and g-C_3N_4-γ-Al_2O_3 (CN-Al) composite catalyst are as follows: Ti-Al was synthesized by impregnating 15 ml titanium butoxide uniformly into 15 g γ-Al_2O_3 together with 20 ml ethanol, followed by drying and calcination at 650 °C for 5 h [29]. 650 °C was chosen as the calcination temperature of catalysts to keep activity of TiO_2 to the greatest extent [30]. While the preparation of CN-Al started from dissolving 10 g melamine into deionized water, after stirring well at 80 °C, 10 g γ-Al_2O_3 were further mixed with the solution and dried at 105 °C overnight. And then the as-received precursor was heated from room temperature to 250 °C and held for 1 h; continuously heated to 350 °C and kept for 1 h; then to 550 °C and held for 4 h under N_2 atmosphere, with a heating rate of 1 °C/min throughout the heating process [21].

Further, $Ni(NO_3)_2{\cdot}6H_2O$ was first dissolved in ethanol followed by mixing with TiO_2, γ-Al_2O_3 and the as-received catalysts (Ti-Al and CN-Al) for the preparation of Ni-based catalysts including Ni-TiO_2 (Ni-Ti), Ni-γ-Al_2O_3 (Ni-Al), Ni-TiO_2-γ-Al_2O_3 (Ni-Ti-Al) and Ni-g-C_3N_4-γ-Al_2O_3 (Ni-CN-Al). Then the mixed solutions were calcined from room temperature to 650 °C with a heating rate of 10 °C/min and a holding time of 2 h after drying overnight. The Ni content was controlled at the amount of around 10 wt% for the Ni-based catalysts by adjusting the amount of $Ni(NO_3)_2{\cdot}6H_2O$ addition. Finally, all catalysts were crushed into 60–150 μm powders for further experiment.

2.2. Plastic photothermal conversion reaction system

Plastic photothermal conversion was conducted in a photothermal catalytic pyrolysis system as shown in Fig. 1. It mainly consists of a N_2 feeding system, a 4 kW solar simulator, a condenser unit and a gas collecting station. The ray from the solar simulator is similar to the concentrated solar radiation (mostly in the visible and IR spectra), with flux densities ranging from 0.2 to 1.5 MW/m^2 within 800 mm diameter circle. Firstly, a focalizer was used to calibrate the position of the reactor to make sure that the sample was located at the focus of the solar simulator. Then, the LDPE feedstock and catalysts were placed in a hanging basket with a mass ratio of 1:2, and loaded into the reactor. The total amount of feedstock and catalysts is around 3 g (±0.2 g) and the exact value is confirmed by the weigh difference of empty basket and basket filled with feedstock and catalysts. Before each experiment, nitrogen was supplied as carrier gas at a flow rate of 200 ml/min and maintained for at least 10 min to make sure the inert atmosphere inside the reactor. The reaction begins once turning on the simulator and setting the current to 50 A (According to our previous calibration results, the temperature of the solar simulator focus at 50 A was 500 °C). The heating rate is around 200 °C/min based on several temperature calibrations before the experiment. The reaction time lasted for 15 min to ensure the complete conversion of feedstock. In addition, air and water cooling were supplied to prevent the overheating of the reactor. The

Table 1
Proximate and ultimate analyses of feedstock.

Feedstock	Ultimate analysis, d%					Proximate analysis, d%		
	C	H	N	S	O	V	A	FC
LDPE	85.13	14.36	0	0	0.51	99.9	0.08	0.03

Note: d-dry basis; V-Volatiles; A-Ash; FC-Fixed carbon.

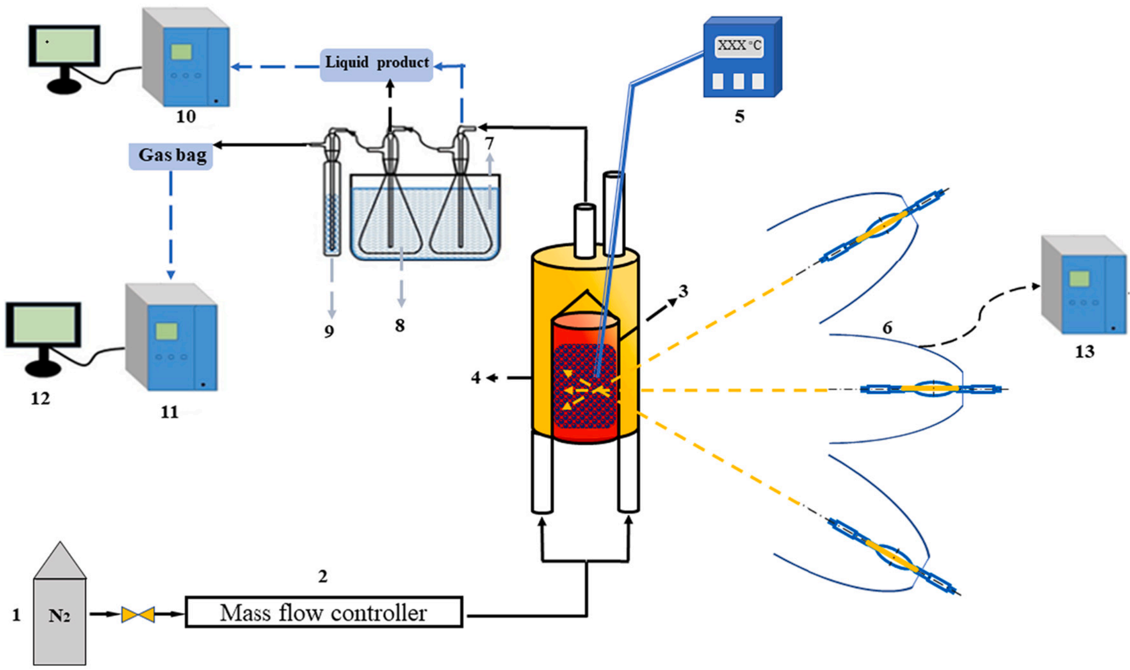

Fig. 1. The photothermal catalytic pyrolysis system.
(1) Nitrogen gas, (2) Gas flow meter, (3) Hanging basket, (4) Quartz reactor, (5) Thermocouple, (6) Solar simulator, (7) Sink, (8) Condenser, 9) Drying tube, 10) GC–MS, 11) GC, 12) PC, 13) Controller.

liquid products were collected by a condenser, the non-condensable gas was collected by a gas sampling bag. Where the solid yield was calculated by the weight difference of the hanging basket before and after the experiment, and the liquid yield was obtained by subtraction of the solid and gas yields.

2.3. Analytical methods

The gas products were determined by a gas chromatograph (GC, A91, PANNA, China), and the composition, as well as the relative concentration, were analyzed by calibrating standard gas mixtures of H_2, H_4, C_2H_6, C_2H_4, C_3H_8/C_3H_6, C_4H_{10}/C_4H_8, CO and CO_2. The organic composition of the liquid product was analyzed by a gas chromatography–mass spectrometry (GC/MS, 7890B/5977A, Agilent, USA) equipped with a DB-5 capillary column. The *m/z* range of MS results was 30–500. Before analysis, the liquid product was collected and diluted with acetone, and the particulate matter in the sample was filtered using a micro filter (aperture 0.22 μm) to avoid damage to the instrument. The spectral data obtained after detection were matched with the spectral data in the standard spectral database NIST2014 to identify the generated compounds, in which the relative peak area was used as a semi-quantitative basis for the content of organic components. The relevant parameters were as follows: oil injection volume was 1 μl, sample injection temperature was 280 °C, the flow rate of carrier gas (He) was 1 ml/min, and split ratio was 20:1.

The specific surface area and pore structure of the composite catalyst were measured by a nitrogen isotherm adsorption desorption analyzer at 77 K. The catalyst samples were first degassed at 200 °C for 3 h to remove water and impurities, and then the test was carried out in liquid nitrogen to obtain the adsorption and desorption isotherms at a constant temperature. The specific surface area was calculated according to the Bruner-Emmett-Taylor (BET) theory, while the micropore volume, micropore and external area were measured by the t-plot method [31]. The morphology of the catalysts was measured by the field emission scanning electron microscopy (FSEM), and the distribution of Ni and Ti species on the surface of the catalysts were identified by the energy dispersion X ray spectra (EDX). The crystal structures and properties of the catalysts were characterized by EMPYREAN X-ray diffractometer (XRD, Panax Analytical B·V) coupled with Cu-K α radiation X-ray source using the scanning range of 2θ = 0–90° [21].

In addition, the carbon deposition on the spent catalyst was analyzed by the temperature programmed oxidation (TPO) to obtain the type and crystallinity quantitatively and qualitatively by studying the changing law of its weight loss with temperature. The TPO analysis was carried out on the PerkinElmer TGA 8000, of which the temperature was programmed to 800 °C at a heating rate of 20 °C/min in air. Furthermore, Raman (Horiba JobinYvon LabRAM HR800) were carried out to investigate the defects and graphitization of deposited carbon.

3. Results and discussion

3.1. Characterization of fresh prepared catalysts

Fig. 2 showed the morphologies of all the fresh synthesized catalysts. Al_2O_3 based catalysts such as Ti-Al and Ni-Al displayed porous and loose structure, while Ni-Ti catalyst showed much dense morphology. It can be seen there was a layer of two-dimensional network, which referred to the g-C_3N_4 species, covered on the CN-Al and Ni-CN-Al. And for Ni-Ti-Al and Ni-CN-Al, the element distribution cloud chart presented that all active sites were uniformly dispersed on the surface. Fig. 3 showed the pore size distributions of the fresh prepared catalysts [32]. The detailed porosity information and pore size of catalysts were summarized in Table 2. All the fresh prepared catalysts showed abundant mesopores, with the average pore size mainly in the range of 7 to 11 nm. CN-Al and Ti-Al had the highest specific surface area of 277.04 m^2/g and 203.62 m^2/g. The introduction of Ni into catalysts lead to a decrease in both specific surface area and pore volume, indicating that the Ni species may block pores inside the structure of the aluminum support.

The XRD patterns of prepared catalysts were shown in Fig. 4. For all Al-based catalysts, there were characteristic diffraction peaks at 37°, 45.5°, 67.5°, representing the existence of γ-Al_2O_3. For Ni-based catalysts including Ti-Al, Ni-Ti and Ni-Ti-Al, the anatase phase of TiO_2 were clearly observed at 25.3°, 37.8°, 48.1°, 53.9° and 55.1°, which corresponded to (101), (004), (200), (105) and (211) planes respectively. It

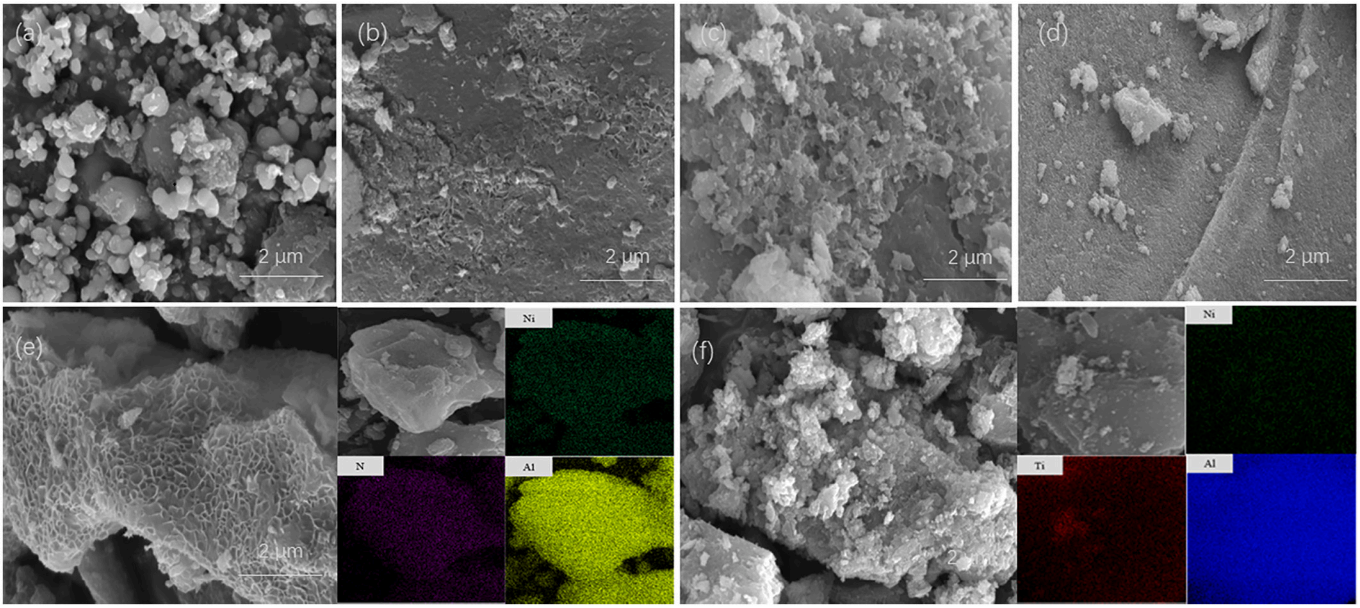

Fig. 2. FSEM images of fresh prepared catalysts, (a) Ti-Al, (b) CN-Al, (c) Ni-Al, (d) Ni-Ti, (e) Ni-CN-Al, (f) Ni-Ti-Al.

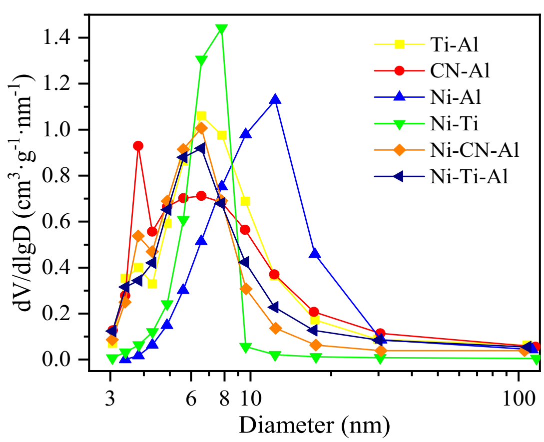

Fig. 3. Pore size distributions of fresh prepared catalysts.

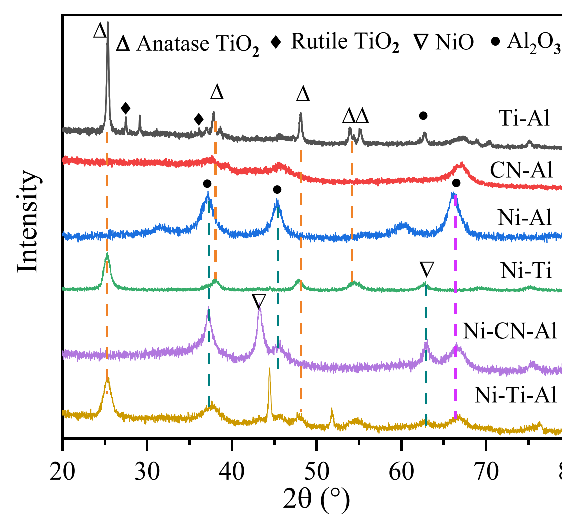

Fig. 4. XRD pattern of pristine catalysts.

Table 2
Porosity analysis of the fresh prepared catalysts.

Catalysts	Specific surface area (m^2/g)	Pore volume (cm^3/g)	Average pore size (nm)
Al	125.06	0.74	23.98
Ti	100.78	0.23	9.42
Ti-Al	203.62	0.50	10.51
CN-Al	277.04	0.52	7.48
Ni-Al	143.97	0.48	14.17
Ni-Ti	138.97	0.29	8.59
Ni-CN-Al	171.96	0.36	8.37
Ni-Ti-Al	183.11	0.41	8.78

should be noted that two weak peaks appeared at 27.4 and 36.1 implied the presence of rutile phase, which was resulted from the anatase phase transformation at calcination temperature above 500 °C [33]. However, the graphitic coating related to g-C_3N_4 was observed in the FSEM images, which can hardly be found in the XRD images of the CN-Al. It may be explained by the low loading amount or incomplete crystalline of thin film structure [21]. For Ni-based catalysts, well resolved peaks of NiO at 37.2°, 43.3°, 62.9° from (111), (200), (220) planes were presented indicating that Ni has been loaded into support materials successfully.

The optical absorbance and band gap of synthesized catalysts were characterized by ultraviolet–visible diffuse reflectance spectroscopy. As showed in Fig. 5, the absorption peak of Ni-Ti-Al had a significantly red shift compared to that of Ti-Al. Ni-Ti-Al not only showed enhanced absorbance in the range of 400–800 nm, but also maintains intense signal at 200–400 nm. Previous research results concluded that low energy photons (>400 nm) were able to drive thermal catalysis, while high energy photons (200–400 nm) can drive photocatalysis, implying integrating potential of Ni-Ti-Al for both thermal catalysis and photocatalysis. It can utilize solar energy more efficiently and save energy cost during the thermo-chemical conversion [34,35]. Ni-CN-Al exhibited the similar variation trend but with lower peak intensity than that of Ni-Ti-Al, indicating that Ni-Ti-Al had a stronger light response capability and higher light absorption efficiency. The band gap energy of catalysts was calculated by Tauc plot method. The band gap energy of Ni-Ti was about 2.88 eV, which was lower than that of intrinsic TiO_2 (3.2 eV). Moreover, Ni-Ti-Al had a narrower band gap energy of 2.69 eV compared with Ni-Ti (2.88 eV). Because γ-Al_2O_3, as a porous support can promote the

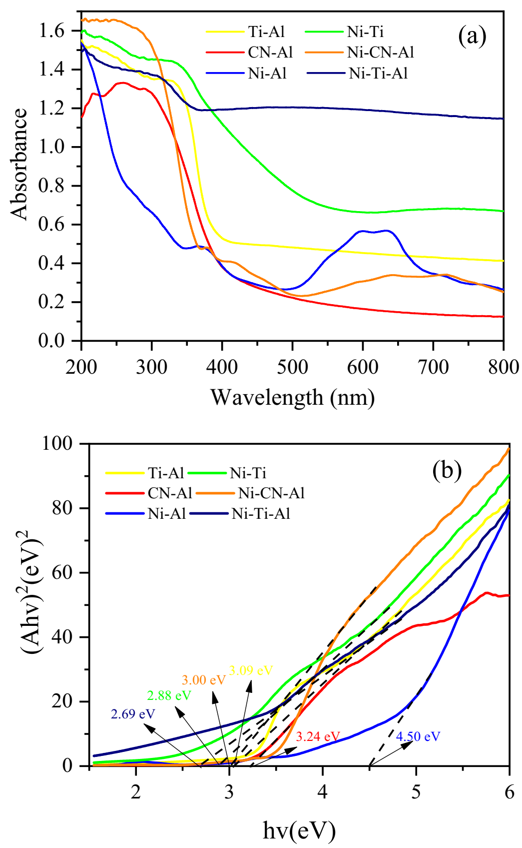

Fig. 5. UV–vis image of prepared catalysts.

dispersion of active component and might adjust the energy band structure to some extent [34].

3.2. Final product distribution

Pyrolysis of LDPE at 500 °C in the absence of catalyst was carried out first and it was found that the condensable product was almost wax with the yield as high as about 90 wt%. This result was consistent with previous studies on the pyrolysis of plastics [6,36]. It is believed that the wax with carbon number larger than 23 made it hard to be further treated or recycled, thus the catalytic trials were then performed, and the products distribution were obtained.

As shown in Table 3, the addition of catalysts all increased the yields of liquid and gas, indicating that they promoted the decomposition of plastic polymers. Ti-Al produced the gas yield of 44.85 wt% and liquid yield of 33.97 wt%, while the addition of Ni showed the decrease in gas yield (28.42 wt%) and increase in liquid yield (48.71 wt%) for Ni-Ti-Al. The same variation trend was found for CN-Al And Ni-CN-Al. This may be ascribed to the active phase Ni, which had high activity to break the C-H and C-C bonds of volatiles for further aromatization and cyclization to form the light hydrocarbon molecules in liquid products, while decreased the gas yield [37]. In addition, there was an increase in the deposition of carbon on the Ni-based catalysts, with the highest amount reaching 45.79 wt% for Ni-Ti. Studies had shown that Ni was easier to cause the coke formation and aggregation of active particles compared to the noble metals, which directly increased the solid product and would further lower the catalytic activity due to the blockage in porous structure and overlay of active sites [38]. This was consistent with the results from BET result shown in Table 2. In addition, Serrano et al. also found that the blocked pore structure, as well as the bulky nature of the polyolefins, improved the probability of the occurrence of steric and diffusional hindrances for entering the catalysts, increasing the deposition of carbon on Ni-based catalysts [39].

Table 3
Effects of catalysts on different product distribution.

	Catalysts					
	Ti-Al	CN-Al	Ni-Al	Ni-Ti	Ni-CN-Al	Ni-Ti-Al
Products yield (wt%)						
Gas	44.85	34.48	11.95	15.09	18.46	28.42
Solid	21.18	21.57	30.35	45.79	33.33	22.87
Liquid	33.97	43.95	57.70	39.12	48.21	48.71

Compared with Ni-Ti, Ni-Ti-Al had less solid products while more liquid and gas products. It indicates that γ-Al_2O_3 as a catalytic support can provide acid sites and promote the dispersion of active component for decreasing coke formation with enhanced catalytic activity [40]. It was noted that the addition of TiO_2 to Ni-Al reduced the solid product from 30.35 wt% to 22.87 wt% and enhanced the gas product from 11.95 wt% to 28.42 wt%. This might be the interaction between Ni and TiO_2, which improved the transfer of carriers and enhanced the assistance of Infrared Radiation (IR) thermal effect [41]. This was consistent with the result of red shift phenomenon shown in Fig. 5. The good performance of Ni-Ti-Al may be also contributed to the strong interaction formed between alkaline site on TiO_2 and the acid site on γ-Al_2O_3, which was conducive to the dispersion of TiO_2 on the carrier. The element distribution of Ni-Ti-Al presented in Fig. 2 can also explain the good dispersion of active component on the aluminum support. While g-C_3N_4 is also alkaline, the synergistic effect is mainly attributed to the stable preservation of g-C_3N_4 nanoparticles on thermally stable acidic γ-Al_2O_3 and its high crystallinity [21]. Compared to Ni-CN-Al, Ni-Ti-Al has more gas yield and less solid yield, indicating that TiO_2 has stronger catalytic activity than g-C_3N_4.

3.3. Pyrolysis gas composition

Fig. 6 showed the effects of different catalysts on the gas products. H_2 was the dominant gas component for all catalysts, and the highest content reached 86.72 vol% for Ni-CN-Al. It can be observed that the addition of Ni has significantly improved H_2 yield from 13.57 wt% to 30.67 wt% for Ni-CN-Al and from 6.91 wt% to 34.20 wt% for Ni-Ti-Al. In addition, the CH_4 and $C_{2\text{–}5}$ gases from Ti-Al and CN-Al had a relatively high content. These were contributed to the relatively low activity of Ti-Al and CN-Al. When Ni was introduced, the cracking of CH_4 and $C_{2\text{–}5}$, as well as the dehydrogenation were promoted, resulting in an increase in H_2 yield shown in Fig. 6(a) [42]. Compared with Ni-Ti-Al, Ni-Ti had a relatively low H_2 yield of 22.81 wt%. This can be ascribed to the porous structure and acid sites of γ-Al_2O_3, which can promote the dispersion of active particles and enhance dehydrogenation [43]. Additionally, compared with Ni-Al, the addition of TiO_2 induced the interaction with Ni, enhancing the mobility of electron-hole carriers from TiO_2 and reducing the carrier recombination rate, which was favorable to redox reaction of intermediate to further increase H_2 yield [18]. The products yield and gas results under pure thermal and photothermal reaction over Ni-Ti-Al catalyst at 500 °C were shown in Fig. S1. It can be observed that the photothermal presented higher gas and solid yield with liquid product decreasing, and it's more conducive to H_2 production.

3.4. Liquid product characterization

Table. 4 and Fig. 7(a) showed the component of liquid product, which was divided into four main categories: C_8-C_{16} aliphatics, C_{17}-C_{23} aliphatics, aromatics(C < 16) and undesirable components (carbon number above 23) according to the chemical composition of jet fuel (C_8-C_{16} aliphatics and aromatics) and diesel fuel (C_8-C_{23} aliphatics and aromatics) [36]. The selectivity of Ti-Al and CN-Al to jet fuel was 38.35% and 35.12%, which increased to 80.27% for Ni-Ti-Al and 73.25% for Ni-CN-Al respectively. Besides, it can be observed from Fig. 7(a) that both

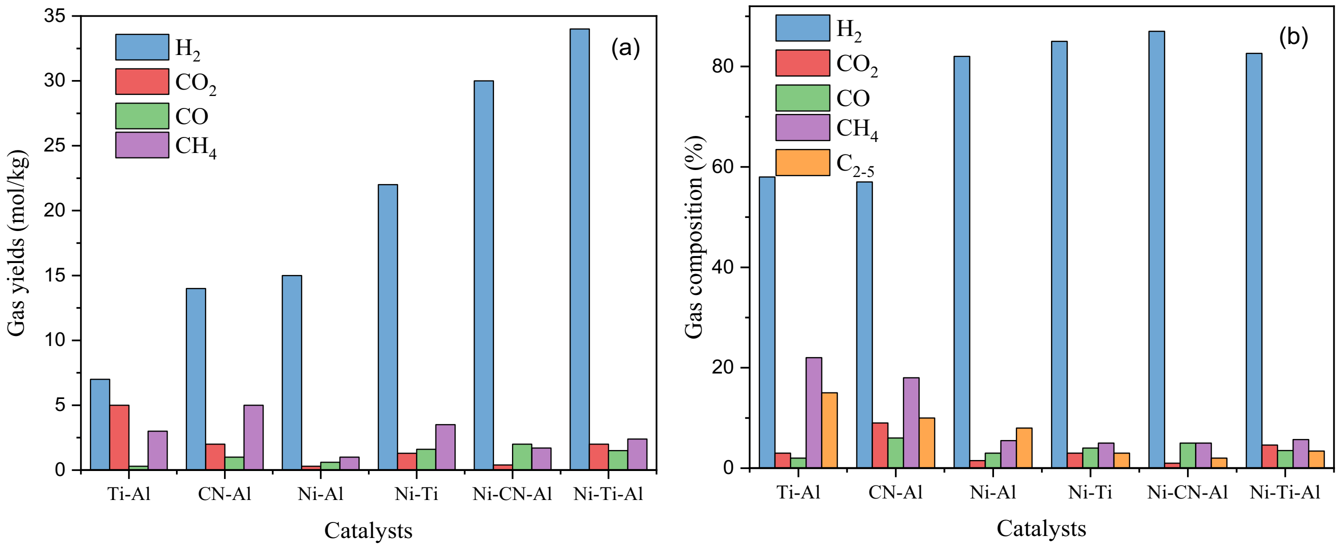

Fig. 6. Effects of catalysts on the gas products of photothermal catalytic pyrolysis of LDPE: (a) gas yields, (b) gas compositions.

Table 4
Liquid product distribution.

	Catalysts					
	Ti-Al	CN-Al	Ni-Al	Ni-Ti	Ni-CN-Al	Ni-Ti-Al
Oil distribution (%)						
Aromatics	13.14	17.75	49.51	31.17	40.25	53.46
Jet fuel	38.35	35.12	68.12	65.01	73.25	80.27
Diesel	78.57	67.23	82.37	85.78	89.25	90.27

Ti-Al and CN-Al had an increased content of aromatics(C < 16) and C_8-C_{16} aliphatics and a reduced content of C_{17}-C_{23} aliphatics after Ni loading. These indicated that Ni promoted the random scission of polymer chain to crack the long chain alkanes into short chain alkanes and enhanced the aromatization and cyclization of light hydrocarbon radicals intermediate to increase aromatics [38]. Miskolczi et al. [44] carried out pyrolysis of waste plastics on a pilot scale reactor with the reaction temperature of 520 °C and feed quantity of 9 kg/h, and the pyrolysis products could be distilled to obtain 20–48% yield of gasoline (C_{5-15}) and 17–36% yield of light oil (C_{12-28}). Artaxe et al. [45] studied the pyrolysis characteristics of HDPE catalyzed by HZSM-5 in a two-stage spout-catalytic bed system, and found that the aromatic hydrocarbon yield could reach 12.5 wt%. Huo et al. [46] investigated the catalytic pyrolysis of LDPE using activated carbon and MgO with 500 °C in a fixed bed thermal reaction system, and found the liquid and gas yield can reach 71.5 wt% and 21.7 wt% respectively, among which C_8-C_{16} and aromatics(C < 16) can reach 58.9% and 18.1% and the H_2 could be up to 92.9%. On the other hand, compared with Ni-Ti-Al, Ni-Ti had a relatively low selectivity to jet fuel, and this can be attributed to the fact that the active particles agglomerated during the reaction and reduced their dispersion, which highlighted the significance of γ-Al_2O_3 support. Compared with Ni-Al, Ni-Ti-Al had higher selectivity for aromatics(C < 16) and jet fuel, which may be explained by the good transfer of carriers between Ni and TiO_2 and localized surface plasmon resonance(LSPR) of Ni with the assistance of IR thermal effect, leading to a unique photothermal synergistic effect in favor of aromatics formation [47,48].

The specific components of aromatic hydrocarbons were further divided into monocyclic aromatic hydrocarbons (benzene, toluene), bicyclic aromatic hydrocarbons (naphthalene, indene), polycyclic aromatic hydrocarbons (anthracene, fluorene) and their derivatives as shown in Fig. 7(b) [49,50]. Ni-Ti-Al had the highest content of 1–2 ring aromatic hydrocarbon of around 33%. This can be ascribed to the aromatization promoted by Ni, which was enhanced by the improved mobility of carriers via the schottky barrier between Ni and TiO_2 [51]. The relative ratio of aliphatic to aromatic were shown in Fig. 7(c). Ti-Al and CN-Al had a dominant aliphatic of 83% and 75% respectively, but the aliphatic of all Ni-based catalysts decreased and had a relatively high amount of aromatics, which was ascribed to the aromatization function of Ni. Among these, Ni-Al has a highest aromatic of 60% followed by 59% of Ni-Ti-Al. The results of liquid composition distribution between pure thermal and photothermal reaction over Ni-Ti-Al at 500 °C were shown in Fig. S2. It can be obviously seen that photothermal improved the aromatic(C < 16) from 38% to 59% with aliphatic decreasing from 62% to 41% comparing with pure thermal.

3.5. Solid product characterization

Considering the relatively high amount of coke deposition and its significant effect on activity of catalysts, several characterization methods were carried out to analyze the types, morphology and graphitization of carbon. The thermogravimetric (TG) and the corresponding differential thermogravimetric (DTG) plots of spent catalysts were shown in Fig. 8. It can be seen that the weight loss curves of six catalysts decreased sharply at 350 °C, which was due to the combustion of deposited carbon materials [52]. All Ni-based catalysts had a large weight loss, it can be obtained that there was a large amount of carbon deposited on these Ni-based catalysts. It can be observed from DTG curves that there were two obvious oxidation peaks located at 350–470 °C and 580–680 °C respectively, which represented two different categories of carbon. Those are amorphous carbon with high oxidation activity at the lower temperature, and filamentous carbon with high graphitization at the higher temperature [53]. Besides, the oxidation peaks of all Ni-based catalysts except Ni-Ti were in the range of high temperature(> 500 °C), which meant carbon deposited on Ni-based catalysts had stronger thermal stability. Ni favored the formation of carbon with high graphitization [54]. The obvious oxidation peak of Ni-Ti was located at 350–470 °C with amorphous carbon deposition. This can be ascribed to the poor morphology of carbon caused by the bad depression of active phase and incidental aggregation. The oxidation peak of Ni-Ti-Al corresponding to the high temperature of 680 °C was the most significant, implying that the largest amount of filamentous carbon was generated. The TPO analysis of spent Ni-Ti-Al between pure thermal and photothermal was shown in Fig. S3. Firstly, the photothermal had a larger weight loss compared with pure thermal, which means more carbon was deposited on catalyst. DTG results show that

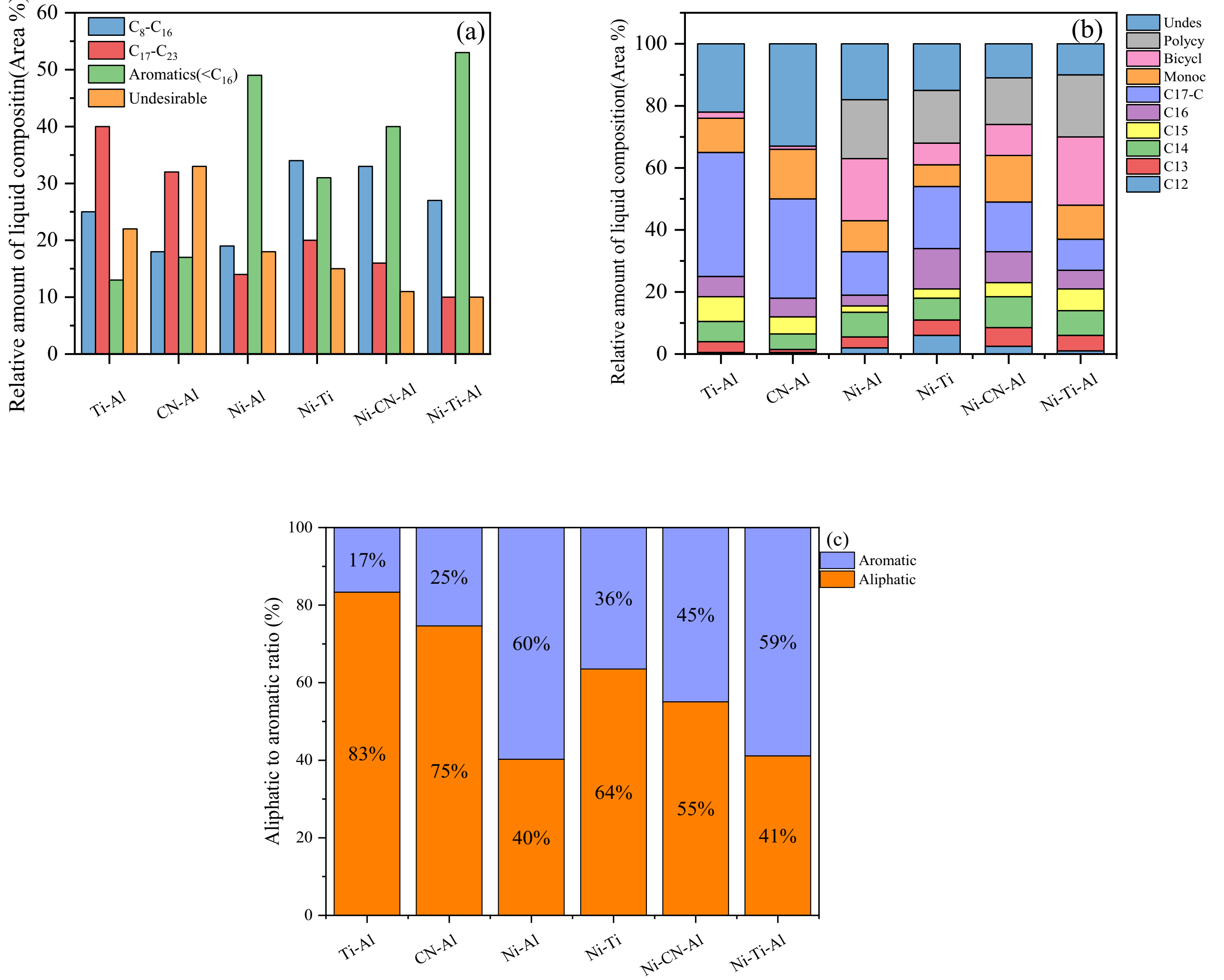

Fig. 7. Effects of catalysts on the liquid products: (a). classified components of liquid product; (b). specific substances of each component; (c). aliphatic to aromatic relative ratio of liquid products.

pure thermal only had an oxidation peak in low temperature range (around 500 °C), but photothermal had another peak above 600 °C, which means photothermal was more beneficial to the formation of filamentous carbon.

The morphology and accumulation status of carbon deposition on catalysts obtained by FSEM was shown in Fig. 9. CN-Al had different sizes of massive carbon block with large and irregular shapes. While the carbon deposited on Ti-Al were in the shape of circular particles with smooth whole, which was relatively uniform. The morphology of carbon deposition changed obviously after Ni loading, a large amount of fibrous carbon can be found especially for Ni-Ti-Al, which was about 40 nm in diameter and several microns in length. Besides, the continuity of carbon tube from Ni-Ti-Al was better, which was ascribed to the fact that the combination of Ni and TiO_2 effectively separated photogenerated carriers and promoted the formation of carbon in a pattern of apical growth [53]. Compared with Ni-Ti, the addition of γ-Al_2O_3 introduced porous structure and acid sites, which resulted in better Ni dispersion and smaller active particles, thus leading to high-quality fibrous carbon tubes in smaller diameter [55]. This result was similar with that obtained by Yeoh et al. [56]. In addition, the agglomeration of large particles was not obviously detected, implying that Ni-Ti-Al occurred less deactivation. SEM analysis of spent Ni-Ti-Al at 500 °C between photothermal and pure thermal was shown in Fig. S4. The obvious difference can be seen that many interwoven fibrous carbons presented on the surface of catalyst after photothermal reaction, while it hardly can be seen on the catalyst after pure thermal reaction.

The graphitization and defect of carbon were further analyzed by Raman spectroscopy and the results were shown in Fig. 10. The D peak and G peak correspond to 1350 and 1580 cm^{-1} respectively, which represent the breathing vibration mode of amorphous carbon on the graphite lamellas and the stretching vibration mode of sp^2 hybrid carbon between the graphite lamellas [54,56,57]. Theoretically, the I_D/I_G value referred to the defects of carbon. It can be seen that the I_D/I_G value of Ti-Al was 0.92, while that of Ni-based catalysts was lower except for Ni-Ti, indicating that Ni can reduce the defect of carbon nanotubes and improve the graphitization. The high I_D/I_G value of Ni-Ti was probably attributed to the fluorescence effect of substrates, causing the intense of I_D not obvious [55]. The I_D/I_G value of Ni-Ti-Al was the smallest (0.84), indicating that the carbon nanotube defects and amorphous carbon were the least, and the graphitization and crystallization degree were the highest. This was consistent with the results of SEM and TPO characterization. In fact, Ni induced hot electrons through its localized surface plasmon resonance (LSPR) under visible light and partially transferred to the TiO_2 surface by plasmon-mediated carrier transfer (PMCT). It

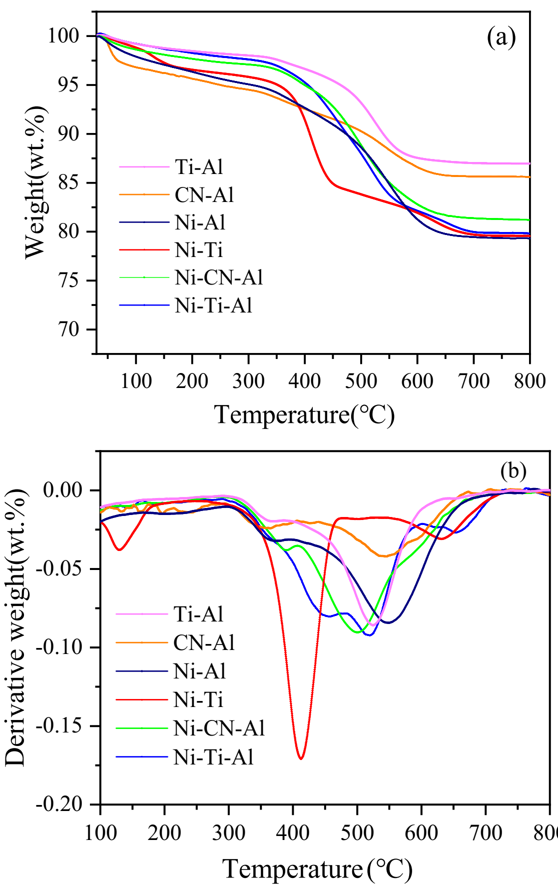

Fig. 8. TPO analysis of spent catalysts: (a). thermogravimetric (TG); (b). differential thermogravimetric (DTG).

produced Ti^{3+} by occupying oxygen vacancies on TiO_2 to effectively extend the life span of hot electrons [34]. And these hot electrons played a dominant role in the formation of fibrous carbon. What's more, TiO_2 induced electron transition effect under UV–vis light, and these electrons were the main driver of carbon nanotubes and benzene rings, which transferred to Ni through the Ni-TiO_2 heterojunction to significantly reduce the recombination of photogenerated carriers [55]. Analogously, for Ni-Ti-Al, the addition of γ-Al_2O_3 also provides acidic sites and loose porous structure, which enhances the dispersion of Ni and TiO_2, promoting the graphitization of carbon.

3.6. Catalytic mechanism

The mechanism of photo-thermal catalysis was shown in Fig. 11. Under the excitation of light, the electrons on TiO_2 transfer from the valence band to the conduction band, thus generating photo-generated electrons in the conduction band and photo-generated holes in the valence band respectively [58,59]. LDPE initially produces long chain hydrocarbons with C-C and C-H bonds by cracking at high temperature [37]. Owing to more electronegative C atoms than H in intermediate molecules, it tends to be adsorbed on the positively charged holes, which is further oxidized and fractured into small molecule hydrocarbons. For

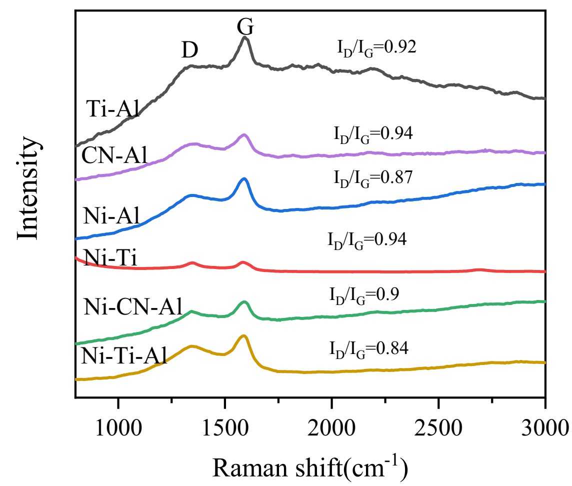

Fig. 10. Raman spectroscopy of carbon deposition on catalysts.

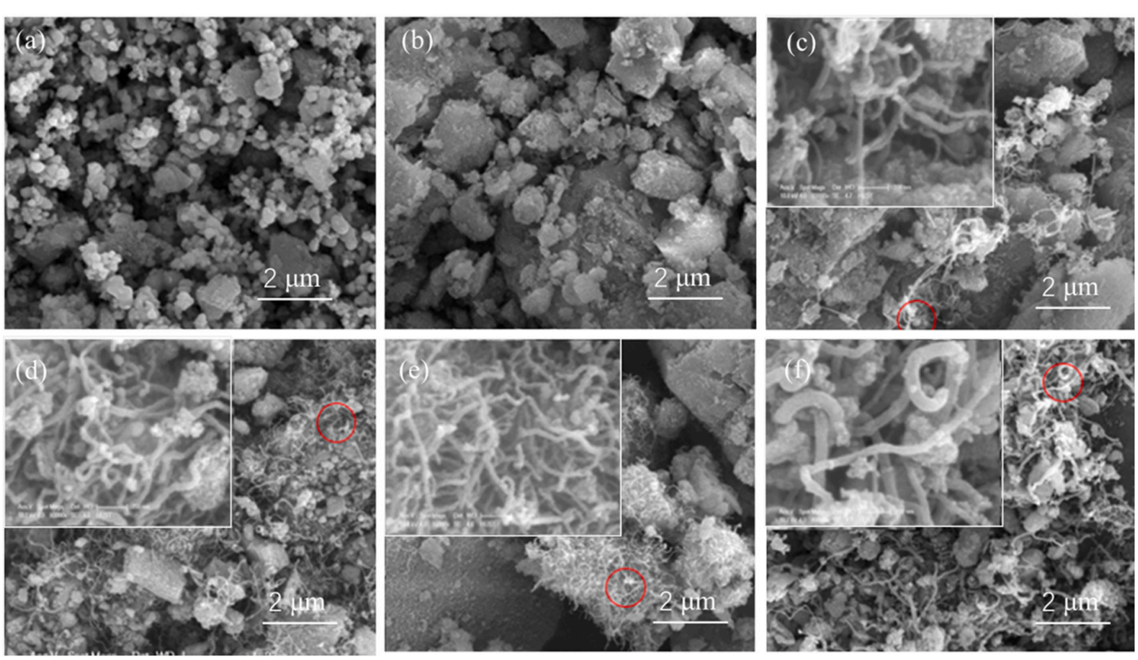

Fig. 9. FSEM image of the spent catalysts (a)Ti-Al, (b)CN-Al, (c)Ni-Al, (d)Ni-Ti, (e)Ni-CN-Al, (f)Ni-Ti-Al.

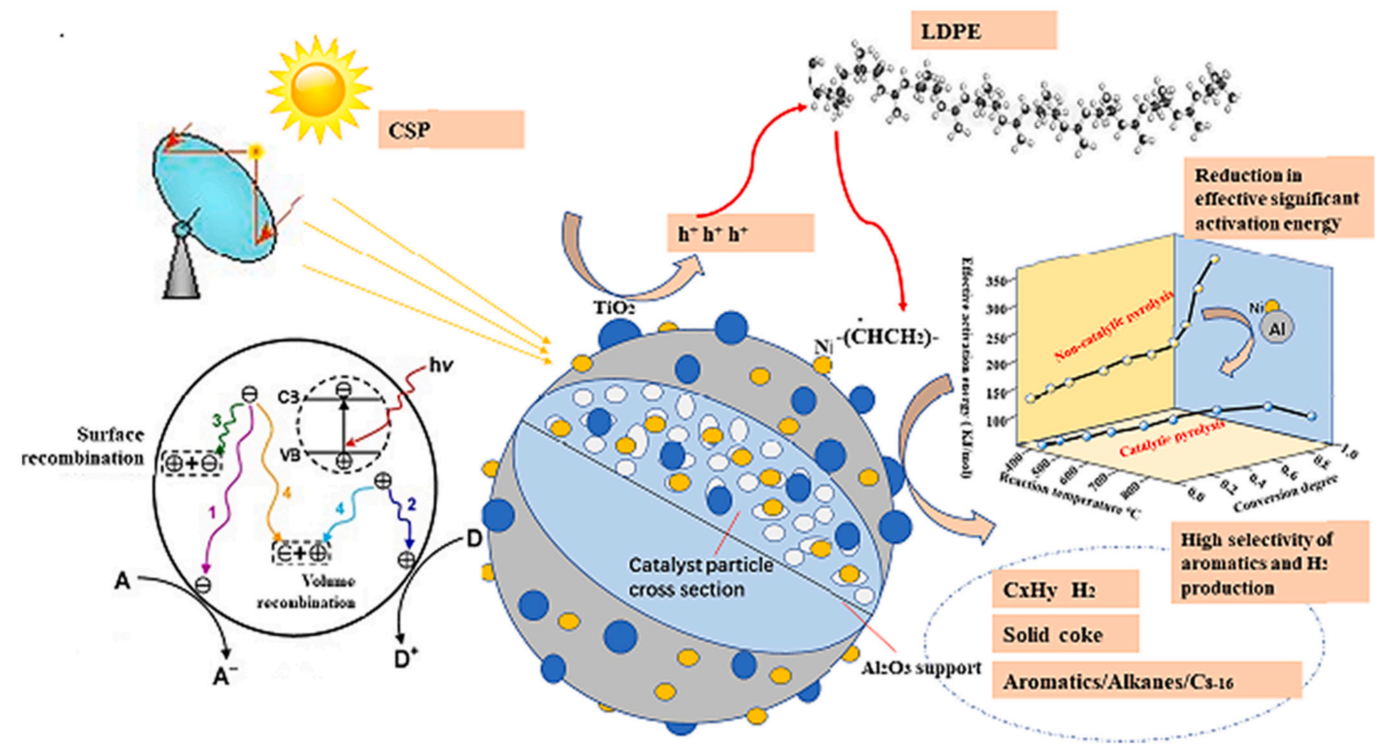

Fig. 11. Mechanism of photo-thermal catalysis.

one thing, loose and porous γ-Al_2O_3 provides a good loading site for the active components, which greatly improves their dispersion. For another, the short chain hydrocarbons are more likely to enter the catalyst through pores and are easier to contact with the active site [60,61]. Ni-based catalysts are more favorable for the generation of high-valued chemicals such as H_2 and jet fuel under photothermal conditions. This is not only ascribed to C-C and C-H cracking function and aromatization of Ni, which is mentioned above, but Ni itself is an excellent photothermal conversion material and can efficiently utilize solar energy. On the other hand, Ni can stimulate localized surface plasmon resonance(LSPR) effect under light condition, which greatly promotes the catalytic activity [51]. At the same time, the heterojunction formed by Ni and TiO_2 is beneficial to the separation of carriers on TiO_2, reducing its recombination rate. In addition, due to Ni doping, the Fermi level of TiO_2 shifts toward the conduction band, which enhanced the absorption of long-wave light and further improved the photocatalytic capacity of TiO_2 [18–20].

4. Conclusion

In this study, six kinds of composite photothermal catalysts supported by Ni, Ti and g-C_3N_4 on γ-Al_2O_3 substrate, were prepared and applied in a novel photothermal reaction system for the pyrolysis of LDPE under 500 °C. The results show that LDPE can produce H_2-rich gas and high added-value liquid products, which is similar to jet fuel after photothermal catalytic pyrolysis. The addition of Ni increased liquid yield from 33.97 wt% to 48.71 wt% and decreased gas yield from 44.85 wt% to 28.42 wt% for Ni-Ti-Al. Among all the catalysts, Ni-Ti-Al displays the most pronounced photothermal synergistic effect, by which LDPE can produce H_2 of 34.20 mol/kg and jet fuel of 80.27% at 500 °C through solar pyrolysis. Compared with Ni-Al and Ti-Al, the ternary-composite catalyst Ni-Ti-Al shows better selectivity for aromatic hydrocarbon composition (53.46%), which has the highest content of 1–2 ring aromatic hydrocarbon of around 33%. Ni-based catalysts show high content of H_2 and aromatics due to its high dehydrogenation and aromatization ability. Ni-Ti-Al also promoted high-quality filamentous carbon with high thermal stability and graphitization degree due to the enhanced carrier transfer though Ni-TiO_2 heterojunction and efficient reduction of carrier recombination rate.

Credit author statement

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Dingding Yao: Formal analysis, Methodology, Writing - Review & Editing, Visualization.
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Haiping Yang: Conceptualization, Resources, Validation.
Hanping Chen: Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuproc.2022.107205.

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